

IT 520 — Which part to improve: find the bottleneck in a kitchen

In-class activity, 20 minutes. Work with the person next to you. Keep this sheet — it is today's participation record.

Tuesday nights at a restaurant, tickets that take **6 minutes** on a quiet evening take **12** during the rush. The head chef already has an answer:

"It's the oven. A tray takes six minutes and everything waits on it. Get me a twice-as-fast oven and this goes away."

A faster replacement oven costs money. Before anyone buys one, do what we did with the server: read the records the kitchen already keeps, work out what one ticket costs each station, find the tallest bar. **You are not timing anything** — both records were already written down.

The ticket log – Tue 18:00-19:00 (the rush)

ticket fired served took
0142 18:03 18:15 12 min
0143 18:05 18:16 11 min
0144 18:07 18:20 13 min
... 12 more tickets in the hour ...

tickets served: 15 average: 12 min

Same log, quiet Tuesday at 17:00:
tickets served: 5 average: 6 min

The station busy-minutes report
– same window, 18:00-19:00 (60 min)

prep busy 45 min 75%
 chopping, portioning, sauces
oven busy 30 min 50%
 5 trays, about 6 min a tray
plating busy 15 min 25%
 the pass

Step 1 — What does one ticket cost each station?

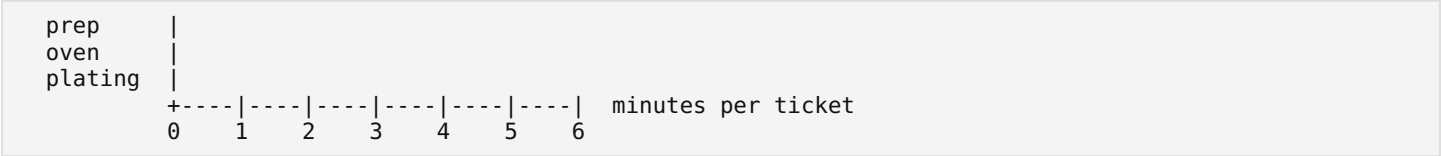
Same division as the disk this morning: $D = B \div C$, busy minutes of a part divided by the jobs that finished. Do it three times.

Station	Busy minutes (B)	$\div$ tickets served (C)	= minutes per ticket (D)
prep			
oven			
plating			

Check yourself before you go on. Add your three numbers: _____ minutes. The log says a ticket takes 6 minutes when nobody is waiting. If those don't match, one of your divisions is wrong — fix it before Step 2.

Step 2 — The tallest bar, and the ceiling

Draw the three bars, then answer.



Bottleneck (the tallest bar): _____ Is it the station the chef named? _____

Ceiling — the station has 60 minutes an hour to give, each ticket takes a bite: $60 \div$ _____ = _____ tickets an hour. Tonight the kitchen served 15.

Step 3 — Price the chef's answer

A twice-as-fast replacement oven halves its service time per ticket. New oven D = _____. Tallest bar now: _____. New ceiling: _____ an hour.

What happened to the ceiling, and why? _____

Step 4 — The change without buying equipment

The vegetables are chopped to order during the rush. Chop them at four o'clock instead and prep takes half as much service time during the rush. No equipment is bought.

New prep D = _____. Tallest bar now: _____. New ceiling: _____ an hour.

The line to take with you: the loudest station is not the bottleneck — the tallest bar is. Only a change that shrinks the tallest bar moves the ceiling, and you verify the change against another supplied record.